

MSc Data Science Project

7PAM2002

Department of Physics, Astronomy and Mathematics

Data Science FINAL PROJECT REPORT

Project Title:

Comparative Analysis of Decision Tree, SVM and XGBoost Models for Heart Disease Risk Classification Using Clinical Features

Student Name and SID:

Hafiz Abdul Razzaq and 23086394

Git Hub Link : <https://github.com/hafiz2343243/Master-Thesis-Heart-Disease-Risk-Prediction-By-using-Machine-learning.git>

Supervisor: [Ralf Napiwotzki](#)

Date Submitted: [21-06-2026](#)

Word Count: [5690](#)

DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at [Assessment Offences and Academic Misconduct](#) and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

I certify that the work submitted is my own and that any material derived or quoted from published or unpublished work of other persons has been duly acknowledged. (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6)

I did not use human participants in my MSc Project.

I hereby give permission for the report to be made available on module websites provided the source is acknowledged.

Student Name printed: Hafiz Abdul Razzaq

Student Name Signature : Hafiz Abdul Razzaq

Student SID number: 23086394

UNIVERSITY OF HERTFORDSHIRE

SCHOOL OF PHYSICS, ENGINEERING AND COMPUTER SCIENCE

Table of Contents

Abstract.....	5
Introduction.....	6
Aim:	6
Research Question:	6
Objective:	6
Literature Review	6
Critical Review of Machine Learning Approaches for Heart Disease Classification	6
Relationship Between Previous Studies and This Project	7
Dataset.....	7
Dataset Source and Purpose	7
Dataset Structure	7
Data Quality Inspection.....	8
Target Variable Creation	8
Exploratory Data Analysis.....	8
Input Features Used for Modelling	10
Ethical Issues.....	10
Use of Public Secondary Data	10
Personal Data and Anonymisation.....	10
GDPR and Data Protection.....	11
Methodology	11
Data Preprocessing.....	11
Train-Test Split and Feature Scaling	11
Model Selection	11
Support Vector Machine Classifier	12
XGBoost Classifier	12
Hyperparameter Tuning	13
Evaluation Metrics.....	13
Results	14
Base Model Results	14
Tuned Model Results	14
Confusion Matrix Results for Base Models	15
Confusion Matrix Results for Tuned Models	16
ROC-AUC Results for Base Models & Tuned Models.....	18
Overall Result Summary	19

Analysis and Discussion	19
Interpretation of Model Performance	19
Effect of Hyperparameter Tuning	19
Comparison with Previous Studies	20
Limitations and Reliability	20
Answer to the Research Question	20
Conclusion	20
Project Summary	20
Main Findings	20
Limitations and Future Work	21
References.....	21

Abstract

Heart disease is an important healthcare problem and early risk classification can help with timely clinical investigation and decision making. This project is about the question of whether Decision Tree, Support Vector Machine and XGBoost models differ in classifying patients into low-risk and high-risk heart disease groups based on clinical health features. The original Kaggle data set had 1025 records and 14 columns. I did data quality checks and found no missing values, and removed 55 duplicate records. I converted the decimal target score to a binary target class using 0.5 as the threshold. The final dataset after the preprocessing contained 970 records and 13 input features. The steps I followed are train test split, StandardScaler, base model training, RandomizedSearchCV tuning and evaluation by accuracy, precision, recall, F1-score, confusion matrix and ROC-AUC. XGBoost had the best base performance and the tuned XGBoost had 1.000 in all metrics. The result indicates effective classification, but external clinical validation is required before any medical use or clinical deployment.

Introduction

Heart disease is an important healthcare problem as it can affect patient well-being, quality of life and long-term survival. Early recognition of risk may facilitate early medical investigation, treatment planning and lifestyle advice. In clinical practice, risk of heart disease is usually estimated from symptoms, medical history, clinical measurements, laboratory results and professional judgement. However, machine learning can be used as a supporting data science approach as structured clinical data may have patterns hard to identify manually.

This project is to classify heart disease risk with clinical health features. This study aims to answer the research question, whether the classification performance of Decision Tree, Support Vector Machine and XGBoost is different in predicting low risk and high-risk heart disease groups from clinical health features. I converted the decimal target score to binary target_class with a threshold of 0.5. The model dataset after pre-processing consisted of 13 input features and 970 records.

Aim:

This research aims to construct, optimize, and assess Decision Tree, Support Vector Machine, and XGBoost models to classify heart disease risk into two categories: low risk and high risk, utilizing clinical health indicators.

Research Question:

Does the classification performance of Decision Tree, Support Vector Machine, and XGBoost vary when predicting low-risk and high-risk heart disease groups from clinical health features?

Objective:

1. Analyze the heart disease dataset to understand its structure, clinical features, target variables, missing values, and duplicate records.
2. Sanitize and preprocess the dataset by removing duplicate entries and converting the original decimal target score into a binary target variable labeled as `target\_class`.
3. Conduct exploratory data analysis to examine the distribution of clinical characteristics and the balance between low-risk and high-risk heart disease groups.
4. Structure the dataset for supervised machine learning by separating the 13 clinical input features from the target variable, using train-test partitioning, and normalizing the input features.
5. Train the Decision Tree, Support Vector Machine, and XGBoost classification models using the supplied training data.
6. Enhance the Decision Tree, Support Vector Machine, and XGBoost models via hyperparameter optimization.
7. Evaluate the baseline and optimized models using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC.
8. Assess the model results to identify the most efficient classifier for evaluating heart disease risk.
9. Evaluate the effectiveness, limitations, and practical implications of the final model for the project's research question.

Literature Review

Critical Review of Machine Learning Approaches for Heart Disease Classification

Mohan, C. (2019) suggested a hybrid random forest with a linear model for heart disease prediction using Cleveland data set. The study demonstrated that machine learning was able to identify clinically useful patterns with an accuracy of 88.7%. The importance of this paper is that it shows the usefulness of predictive modelling for cardiovascular disease classification. However, it does not match directly my implementation

since I don't use Random Forest or a linear model in my code. In my project, I am comparing Decision Tree , SVM and XGBoost as different classifiers and then tuning for performance improvement.

Ghosh et al. (2021) introduced a cardiovascular disease prediction framework using combined datasets from Cleveland, Long Beach VA, Switzerland, Hungarian and Statlog. Preprocessing, Relief and LASSO feature selection and bagging and boosting classifiers were used in the study. The best result was 99.05% Accuracy with Random Forest Bagging Method using Relief selected features. This paper is relevant to my project as preprocessing, feature relevance and model comparison are important in clinical classification. But their combined dataset and feature-selection pipeline is more complex than my project which uses one prepared dataset, scaling, tuning and three selected classifiers.

L. Fitriyani (2020) proposed an effective heart disease prediction model for a clinical decision support system. The method used DBSCAN for outlier removal, SMOTE-ENN for class balancing and XGBoost for prediction. The study reported 95.90% accuracy on Statlog and 98.40% accuracy on Cleveland. This paper is strongly aligned with my project because XGBoost is one of my implemented models, and the study also compares XGBoost with SVM and Decision Tree. The limitation is that my code does not apply DBSCAN or SMOTE-ENN, so comparison must recognise preprocessing differences.

P. Li (2020) proposed a method for detecting heart diseases using Cleveland dataset. In the study, methods such as SVM, Logistic Regression, Artificial Neural Network, K-Nearest Neighbour, Naive Bayes and Decision Tree were used with feature selection methods such as Relief, MRMR, LASSO, LLBFS and FCMIM[16]. The best results were obtained using FCMIM with SVM classifier. This paper is important because I implemented and evaluated both SVM and Decision Tree, two exact models. But the paper is more on feature selection. My project is more on direct comparison and tuning models with Decision Tree, SVM and XGBoost.

Relationship Between Previous Studies and This Project

The studies reviewed are relevant because they use supervised machine learning for heart disease prediction using clinical data. But my project is a different size. Mohan et al. (2019) used a hybrid Random Forest with a linear model, my code does not use either of them. Ghosh et al. (2021) used a combined dataset and Relief and LASSO feature selection. My project uses one prepared dataset. Fitriyani et al. (2020) used DBSCAN, SMOTE-ENN, and XGBoost while my code used XGBoost without outlier removal or class balancing. Li et al. (2020) used SVM and Decision Tree with feature selection, I use directly the two for tuning and comparison. My project uses 970 cleaned records in total, 13 clinical features and 3 models: Decision Tree, SVM and XGBoost. So my model choices are supported by literature, but the literature also shows that results should be evaluated by several metrics, and not just accuracy.

Dataset

Dataset Source and Purpose

The dataset for this project is taken from Kaggle : Heart Attack Prediction by Juledz The code used the heart.csv file before pre-processing which had 1025 records and 14 columns. It has clinical variables like age, sex, type of chest pain, blood pressure, cholesterol, fasting blood sugar, maximum heart rate, oldpeak, major vessels and thalassemia. Source: Kaggle, original collector, country and collection year not clearly stated. The dataset was chosen for supervised risk classification.

Dataset Link: <https://www.kaggle.com/datasets/juledz/heart-attack-prediction>

Dataset Structure

The initial data set comprised 1025 records and 14 fields. The input variables were clinical and diagnostic health features and the original target column was given as a decimal risk score. After playing around with

the dataset I observed that the column names were cleaned to a common format. Spaces were removed, and the names converted to lowercase. This simplified processing of the data in Python and decreased the chance of programming errors during the model development.

Data Quality Inspection

Prior to performing the modeling, I checked for missing values and duplicate records in the data set. The dataset has not missing values; therefore, no imputation was needed. I removed 55 duplicate records. After de-duplication, the dataset consisted of 970 records. The importance of this is the fact that duplicated rows can artificially boost the model performance especially when similar records exist in both the training and test datasets.

Target Variable Creation

The original target column was a decimal score ranging from 0.1 to 0.9. I created a new target variable called `target_class` to make the project work for binary classification. Low risk was defined as a score less than 0.5, high risk as a score of 0.5 or greater . After this conversion, the dataset contained 477 low risk and 493 high risk records. The distribution was pretty balanced so didn't use any oversampling method like SMOTE in the code.

Exploratory Data Analysis

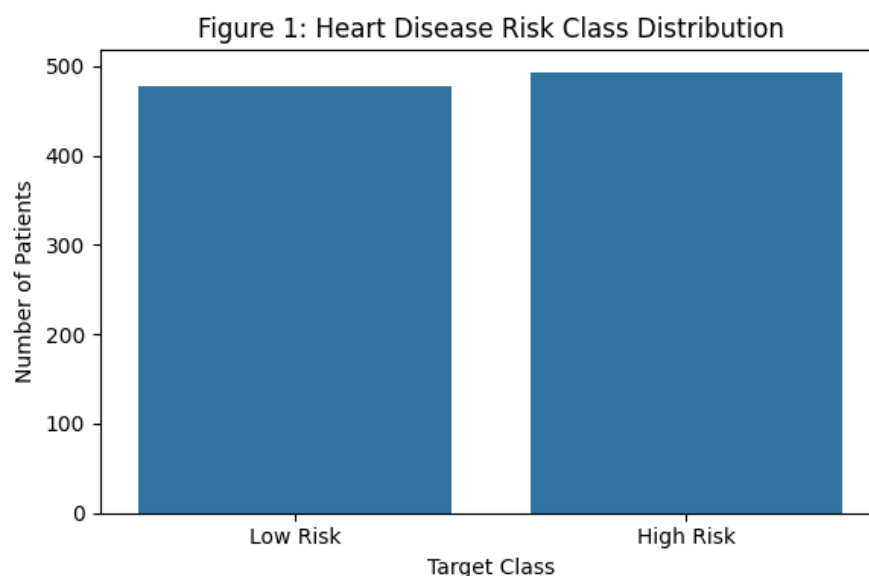


Figure 1: Distribution of Heart Disease Risk Classes

The distribution of the created variable `target_class` is shown in Figure 1. The dataset had 477 low-risk records and 493 high-risk records, which showed that the two classes are relatively balanced. This was useful for modelling as a highly imbalanced target can cause classifiers to be biased towards the majority class. Given the closeness of the distribution, the selected models were trained without the use of oversampling methods like SMOTE.

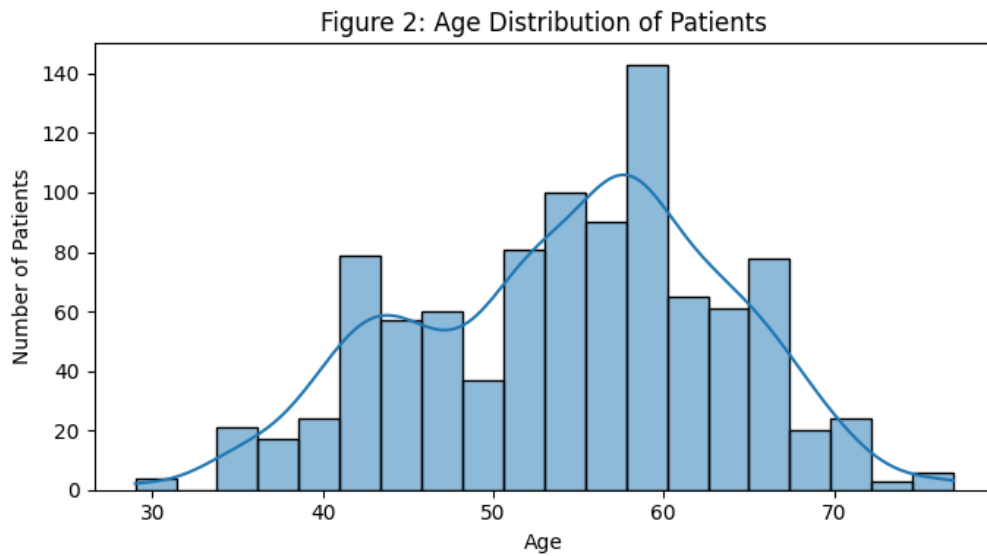


Figure 2: Age Distribution of Patients

Figure 2 shows age distribution of the patients. Most records were concentrated between about 50 and 65 years, although younger and older patients were also present in the dataset. This shows that the dataset has a quite useful age range for heart disease risk classification.

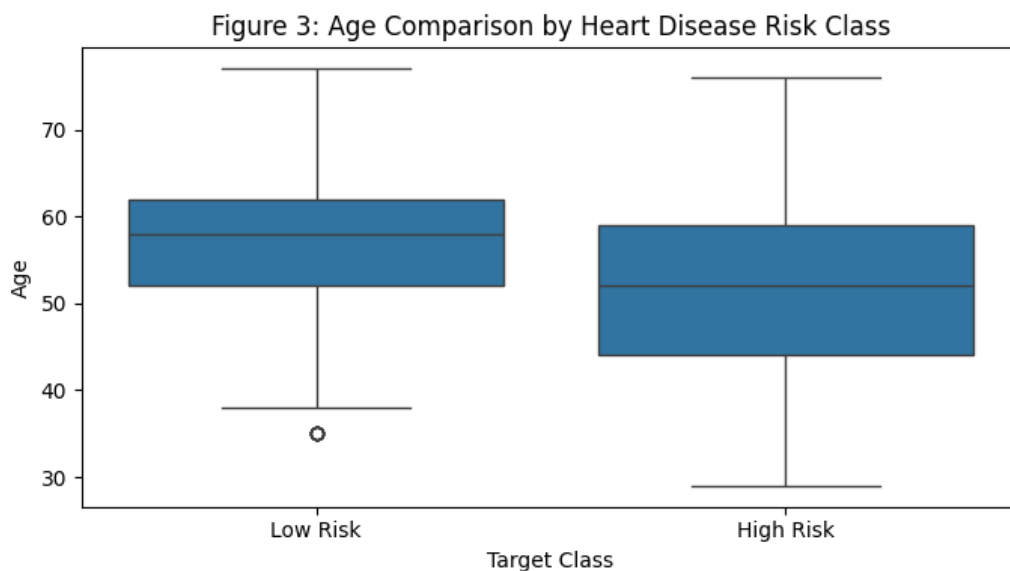


Figure 3: Age Comparison Between Low-Risk and High-Risk Patients

Figure 3: Age of patients in low-risk and high-risk groups. The boxplot shows that the two classes overlap, i.e., that age alone is not sufficient to separate the risk groups. However, the difference in class distributions suggests that age may still be informative when combined with other clinical variables.

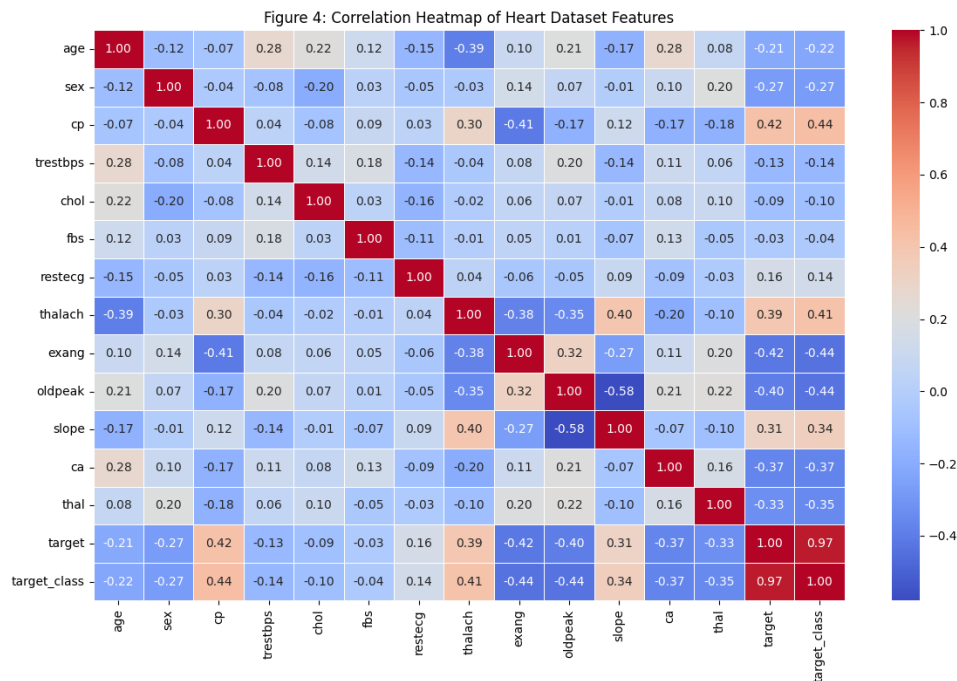


Figure 4: Correlation Heatmap of Clinical Features and Target Class

The correlation heatmap is shown in Figure 4. The variables that had the strongest relationships with target_class were chest pain type, maximum heart rate, exercise induced angina, oldpeak, slope, number of major vessels and thalassemia . But target_class was derived from the original target column, so the original target column had a very strong correlation with target_class. Therefore, target and target_class were removed from the input feature set before modelling to prevent data leakage. The overall EDA confirmed that the data set was suitable for supervised classification with Decision Tree, SVM and XGBoost.

Input Features Used for Modelling

For modelling I excluded the original target column and the newly created target_class column from the input feature set. This ensured that the models were only trained on the clinical input variables and were not provided with the answer column during training. The final feature matrix consisted of 13 clinical variables which were used to train and test the three selected machine learning models: Decision Tree, SVM and XGBoost. The final modelling dataset was thus directly aligned with the project aim of classifying patients into low-risk and high-risk heart disease groups based on clinical health features.

Ethical Issues

Use of Public Secondary Data

For this project I have used a heart disease dataset which is available publicly on Kaggle. The dataset was used only for academic analysis and development of machine learning models. I did not gather new data from individuals, administer surveys, interview participants, reach out to patients or request anyone to test a system. Thus, there was no direct human involvement in the project.

Personal Data and Anonymisation

The dataset is health-related but there are no direct personal identifiers such as names, addresses, telephone numbers, email addresses or hospital identification numbers. The available columns contain clinical and demographic variables such as age, gender, type of chest pain, blood pressure, cholesterol and maximum heart rate. Thus individual patients cannot be directly identified from the dataset used in the project.

GDPR and Data Protection

GDPR and data protection issues were considered in the project, as health-related data is sensitive. The risk of identifying individuals is low as the dataset is secondary public data and does not contain direct identifiers. The data was stored and used for this academic project only and no attempt was made to identify any individual patient.

Methodology

Methodology was carried out in the same manner as the practical workflow of the code. Loaded heart disease dataset, checked the structure, cleaned the column names, checked for missing values and removed 55 duplicate records. I changed the original decimal target score into the binary target_class. Low risk was defined as a score of less than 0.5 and high risk a score of 0.5 or more. I removed target columns from input features, split data into train and test sets and applied StandardScaler. Finally, I trained Decision Tree, SVM and XGBoost, tuned it using RandomizedSearchCV and evaluated performance using six metrics for fair overall comparison.

Data Preprocessing

Data preprocessing was completed before model training to prepare the dataset for supervised machine learning. I normalized the column names by removing spaces, converting text to lowercase, and replacing spaces with underscores. This improved consistency while utilizing the dataset in Python. I subsequently investigated for missing data and duplicate entries. No missing data were detected; hence, imputation was unwarranted. Fifty-five duplicate records were removed, reducing the dataset from 1,025 to 970 records. I converted the decimal goal score into a binary target_class, where values below 0.5 signify low risk and values of 0.5 or more denote high risk.

Train-Test Split and Feature Scaling

Train-Test Split

After preprocessing, I split the data set into training and testing sets and separated the 13 clinical input features from the target variable. The training set contained 776 and the testing set contained 194 records. This split was important because the models had to learn the patterns from the training data and then test unseen testing data. I used a separate test set to see how well the models could generalize outside the training data.

Feature Scaling

The features were scaled using StandardScaler. The scaler was fit on the training data and then used on both the training and testing data to prevent data leakage from the test set. This standardises the input features so that variables with larger numerical ranges (e.g. cholesterol and blood pressure) do not dominate variables with smaller numerical ranges. Scaling was particularly important for SVM, since it is sensitive to the magnitude of the features. Then the scaled data was used to train and evaluate Decision Tree, SVM, and XGBoost consistently.

Model Selection

Decision Tree Classifier

Decision Tree is a supervised machine learning model that makes predictions by splitting the dataset into branches depending on feature values. Each split produces a decision rule, and the final prediction is made at a leaf node. In classification, the model assigns each record to a class depending on the decision path taken through the tree.

$$Entropy(S) = - \sum_{i=1}^c p_i \log_2(p_i)$$

Where:

- S is the dataset at a node
- p_i is the proportion of samples belonging to class (i)
- (c) is the number of classes

Why This Model Was Used

A simple and interpretable Decision Tree model was used as a baseline classifier. For this project it was useful to test for clear rule-based patterns in the clinical features to distinguish the low and high risk groups of heart disease. It also served as a useful benchmark for the more complicated SVM and XGBoost models.

Base parameters used

DecisionTreeClassifier(random_state=42)

Tuned parameters

criterion = gini, max_depth = 8, min_samples_split = 5, min_samples_leaf = 1

Support Vector Machine Classifier

The Support Vector Machine is a supervised classification model that finds the optimal separating hyperplane between classes. The model aims at creating a boundary with the maximum margin between the classes. This makes it useful for binary classification tasks where the aim is to divide two groups.

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

Where:

- (w) is the weight vector
- (b) is the bias term

Why This Model Was Used

SVM was used because this project is a binary classification task, classifying patients into high-risk and low-risk groups. It was appropriate to see if the scaled clinical features could provide a clear separation of the two classes. Because SVM is sensitive to feature magnitude, StandardScaler was applied before training to make the comparison between models fair and consistent.

Base parameters used

SVC(probability=True, random_state=42)

Tuned parameters

C = 10, kernel = rbf, gamma = scale, degree = 2

XGBoost Classifier

XGBoost is a supervised boosted tree-based classification model. It builds a number of decision trees in a sequential manner, each new tree trying to fix errors of previous trees. This boosting procedure improves the predictive performance of the model on structured tabular data.

$$Prediction_{new} = Prediction_{old} + \eta \times Tree_{output}$$

Where:

- ($Prediction_{old}$) is the previous model prediction
- (η) is the learning rate

- ($Tree_{output}$) is the prediction made by the next decision tree

Why This Model Was Used

XGBoost was chosen because it is appropriate for structured clinical data and is a common approach to machine learning classification problems. This was incorporated into this project to see if a boosted tree-based model could perform better than the simple Decision Tree and the margin-based SVM model. This enabled the identification of the most powerful classifier for the created target_class.

Base parameters used

`XGBClassifier(random_state=42, eval_metric="logloss")`

Tuned parameters

`n_estimators = 200, learning_rate = 0.1, max_depth = 6, subsample = 1.0, colsample_bytree = 0.7`

Hyperparameter Tuning

Hyperparameter optimization was conducted subsequent to the training of the foundational models. The optimization sought to enhance model performance by evaluating various parameter combinations and identifying the optimal configurations for each classifier. This project employed RandomizedSearchCV for the Decision Tree, SVM, and XGBoost algorithms. I employed RandomizedSearchCV due to its superior efficiency compared to exhaustively assessing all parameter combinations. RandomizedSearchCV is effective for assessing several parameters. It methodically examines a specified range of hyperparameter values supplied by the user and employs cross-validation to assess the optimal combination.

The general tuning process can be represented as:

$$\theta^* = \arg \max_{\theta \in \Theta} CV(\theta)$$

Where:

- θ^* represents the best hyperparameter combination
- (Θ) represents the set of possible hyperparameter values
- $(CV(\theta))$ represents the cross-validation score for a specific parameter setting

Evaluation Metrics

The model performance was assessed using accuracy, precision, recall, F1-score, confusion matrix and ROC-AUC. The reason of choosing these metrics is because this project is a binary classification task where patients are classified into two groups, low risk heart disease and high risk heart disease. Accuracy alone may not tell the whole story in terms of how well a model is identifying each class. Using more than one metric is important.

Accuracy is the overall percentage of correct predictions calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision measured how many predicted high-risk cases were actually high risk:

$$Precision = \frac{TP}{TP + FP}$$

Recall measured how many actual high-risk cases were correctly identified:

$$Recall = \frac{TP}{TP + FN}$$

F1-score combined precision and recall into one balanced metric:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

In these equations, TP is true positive, TN is true negative, FP is false positive and FN is false negative. For each model these values were displayed in a confusion matrix which allowed for the identification of the types of classification errors. I used the ROC-AUC to evaluate the ability of each classifier to distinguish between low and high risk classes at different decision thresholds. These metrics were used for both base and tuned models so that Decision Tree, SVM and XGBoost could be fairly compared.

Results

Base Model Results

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Decision Tree	0.835052	0.894118	0.767677	0.826087	0.932111
SVM	0.943299	0.931373	0.959596	0.945274	0.985965
XGBoost	0.969072	0.979381	0.959596	0.969388	0.995747

Table 1: Base Model Performance Results

The base results show that XGBoost achieved the best overall performance before tuning. It had the highest accuracy, precision, F1-score, and ROC-AUC. SVM also performed well especially on the recall metric meaning that a high proportion of true high risk cases were found. The base performance was the lowest in the Decision Tree model when compared with SVM and XGBoost models, but it still achieved a good ROC-AUC score. The results of the base model indicate that the clinical features have useful prediction patterns for classifying low-risk and high-risk heart disease groups.

Tuned Model Results

Base models were evaluated and then hyperparameters of Decision Tree, SVM and XGBoost were tuned using RandomizedSearchCV. Tuning to find the best parameter setting for each model and to compare performance increase after optimization.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Tuned Decision Tree	0.989691	1.000000	0.979798	0.989796	0.999575
Tuned SVM	0.994845	0.990000	1.000000	0.994975	1.000000
Tuned XGBoost	1.000000	1.000000	1.000000	1.000000	1.000000

Table 2: Tuned Model Performance Results

The tuned results show that all three models have improved after tuning the hyperparameters. Tuned Decision Tree did very well with perfect precision and very high ROC-AUC score. The tuned SVM achieved perfect recall and ROC-AUC, correctly identifying all the actual high-risk cases in the test set. The best overall result was achieved by Tuned XGBoost with 1.000000 for accuracy, precision, recall, F1-score and ROC-AUC. Hence, Tuned XGBoost was the best performing model after tuning.

Confusion Matrix Results for Base Models

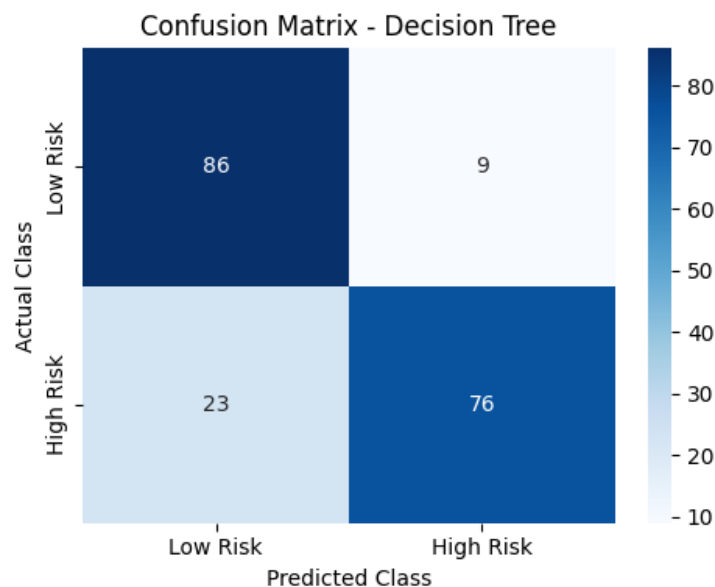


Figure 5: Base Decision Tree Confusion Matrix

The Decision Tree model in Figure 5 had the least base performance. It accurately predicted 86 low-risk patients 76 high-risk patients. But it also produced 9 false positives and 23 false negatives. The high number of false negatives means that the model missed more high-risk patients than the other base models, which is an important limitation for a heart disease risk classification task.

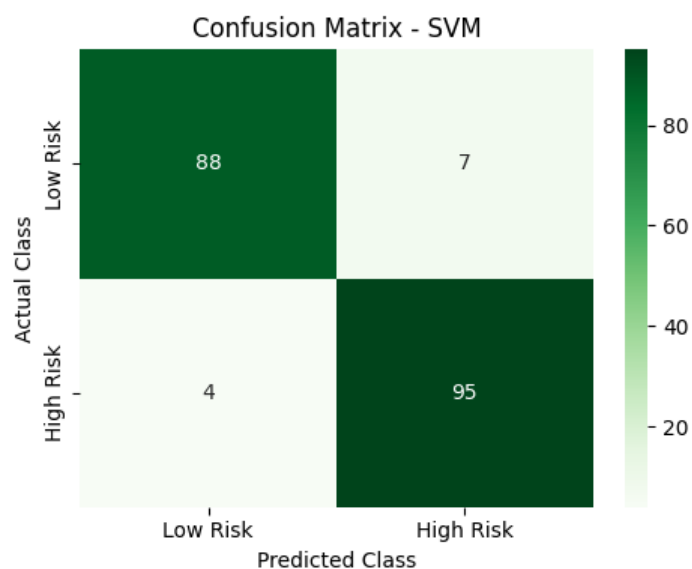


Figure 6: Base SVM Confusion Matrix

The better result was obtained with the SVM model in Figure 6. It was able to identify 88 low-risk patients and 95 high-risk patients. For example, the SVM reduced the number of false negatives from 23 to 4 compared to the Decision Tree, indicating that SVM was much better at identifying high risk patients. but it gave 7 false positives.

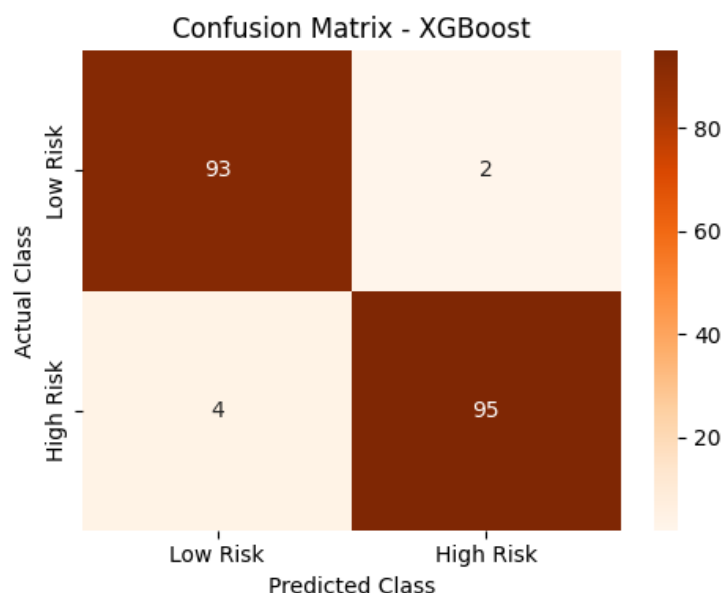


Figure 7: Base XGBoost Confusion Matrix

The XGBoost model shown in Figure 7 obtained the best base confusion matrix. It classified 93 low risk patients and 95 high risk patients correctly with only 2 false positives and 4 false negatives. This means, before tuning, XGBoost had the least total errors. These results support the base performance table, where XGBoost also had the highest accuracy, F1-score and ROC-AUC before hyperparameter tuning.

Confusion Matrix Results for Tuned Models

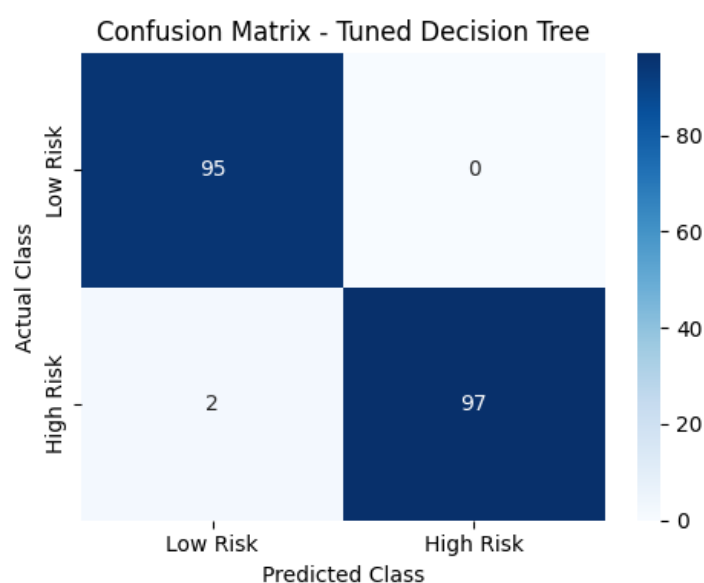


Figure 8: Tuned Decision Tree Confusion Matrix

Figure 8 depicts that the tuned Decision Tree correctly classified 95 low-risk patients and 97 high-risk patients. The model had no false positives and only 2 false negatives. This shows a definite improvement over the base Decision Tree, with the number of missed high-risk cases significantly reduced post tuning.

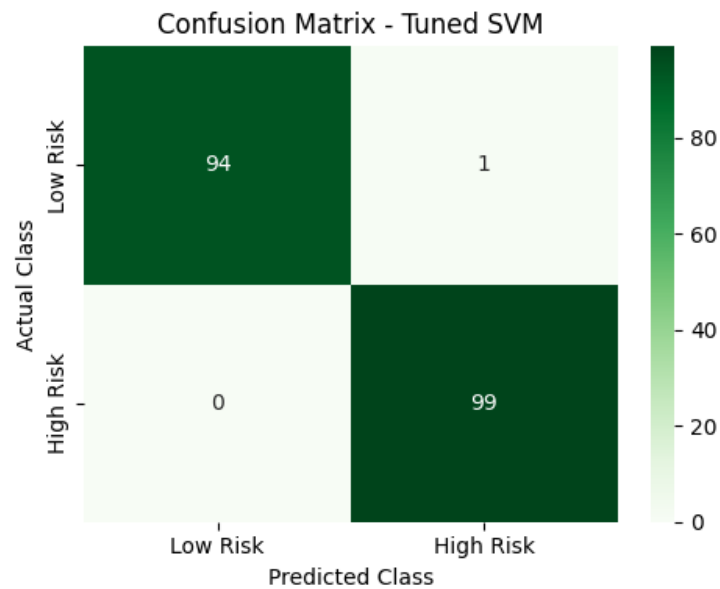


Figure 9: Tuned SVM Confusion Matrix

Figure 9 The tuned SVM correctly classified 94 low-risk patients and all of the 99 high-risk patients. The model produced only 1 false positive and 0 false negatives. The results suggest that the tuned SVM was very effective in identifying true high-risk patients.

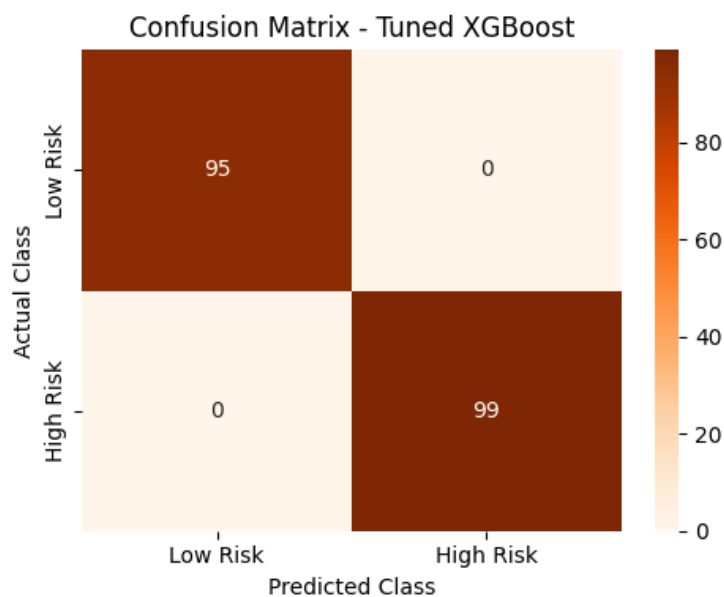


Figure 10: Tuned XGBoost Confusion Matrix

The confusion matrix from the tuned XGBoost model performed best as shown in figure 10. It correctly classified 95 low risk patients and 99 high risk patients, with no false positives and no false negatives. This means that all test records were correctly classified. The tuned confusion matrices, in general, show that the classification performance has improved with hyperparameter optimisation. The tuned XGBoost achieved the best result among the three tuned models.

ROC-AUC Results for Base Models & Tuned Models

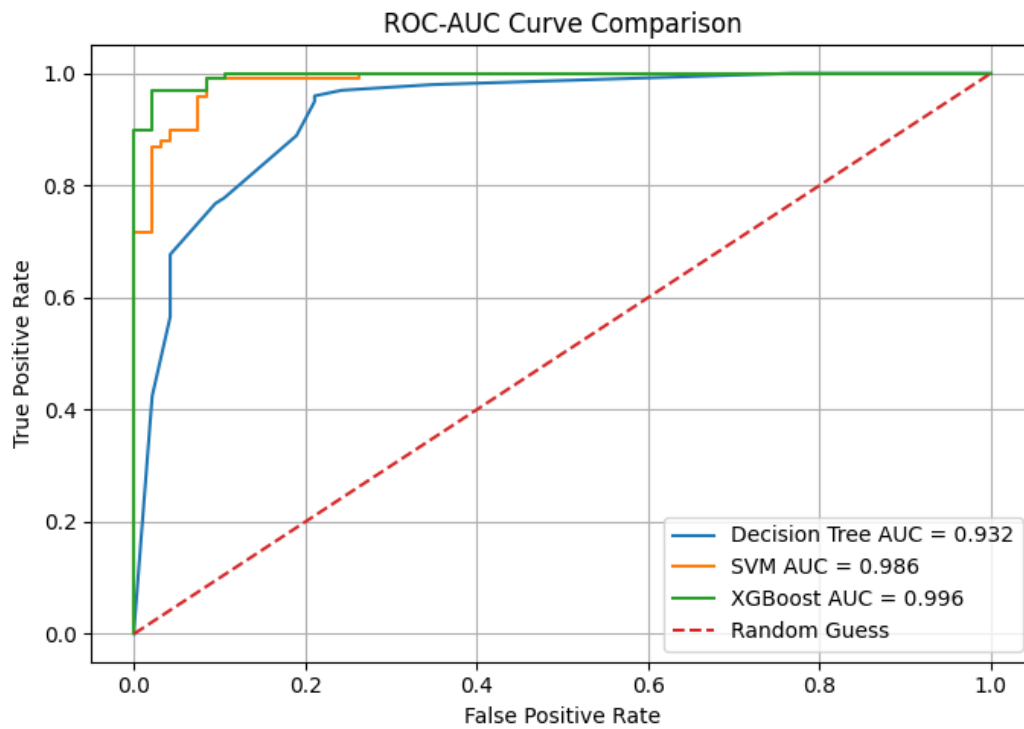


Figure 11: Base Model ROC-AUC Curve Comparison

Figure 11 shows the ROC-AUC curves of the base Decision Tree, SVM and XGBoost models prior to the tuning. The red diagonal line is the random classification and the closer the curves are to the top-left corner, the better the separation of classes. Decision Tree got 0.932. SVM got 0.986 and XGBoost got the highest AUC of 0.996. This confirms that the base discrimination between low and high risk classes was best provided by XGBoost, followed by SVM and Decision Tree. This is consistent with the base performance results in Table 1.

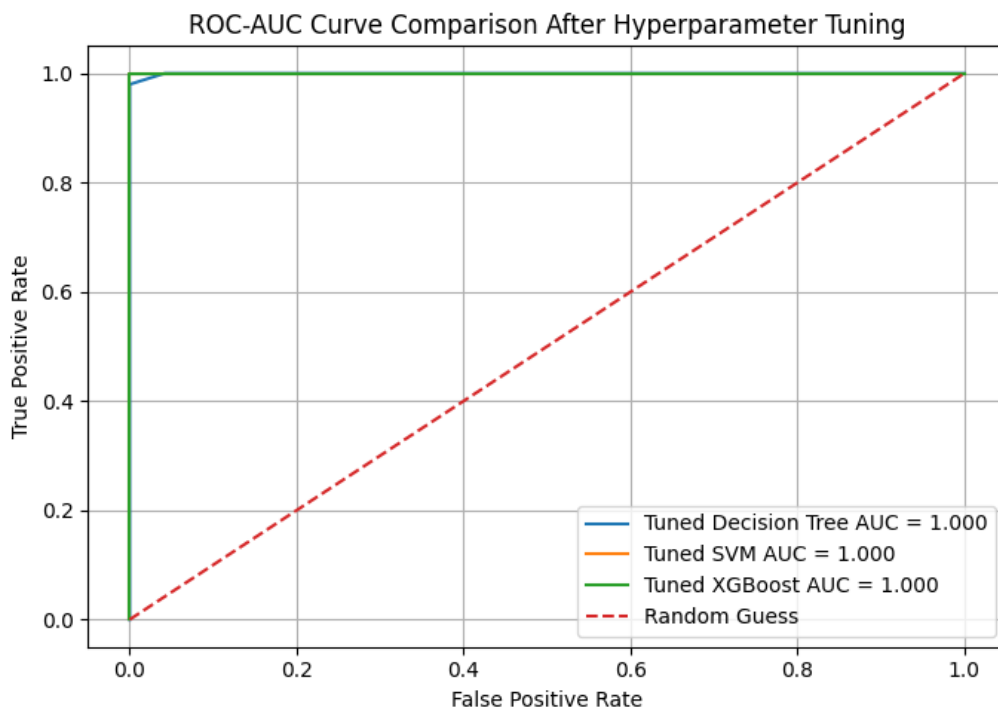


Figure 12: Tuned Model ROC-AUC Curve Comparison

Figure 12 depicts the ROC-AUC comparison following hyperparameter adjustment. The red diagonal line represents random classification, whereas the calibrated model curves are positioned in the top-left corner,

signifying strong class separation. The Tuned Decision Tree achieved an AUC of 1.000 in the presented figure, indicating near-perfect discrimination. The optimized SVM achieved an AUC of 1.000. The optimized XGBoost achieved an AUC of 1.000, matching the pinnacle of performance results. Optimized models markedly exceed random classification across decision thresholds, enhancing separation performance for both low-risk and high-risk patient categorization.

Overall Result Summary

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Base Decision Tree	0.835052	0.894118	0.767677	0.826087	0.932111
Base SVM	0.943299	0.931373	0.959596	0.945274	0.985965
Base XGBoost	0.969072	0.979381	0.959596	0.969388	0.995747
Tuned Decision Tree	0.989691	1.000000	0.979798	0.989796	0.999575
Tuned SVM	0.994845	0.990000	1.000000	0.994975	1.000000
Tuned XGBoost	1.000000	1.000000	1.000000	1.000000	1.000000

Table 3: Overall Base and Tuned Model Performance Comparison

Following hyperparameter optimization, all models exhibited enhancement. The optimized Decision Tree diminished prediction errors, the optimized SVM accurately identified all high-risk patients, and the optimized XGBoost attained the most favorable overall outcome. The optimized XGBoost achieved an accuracy, precision, recall, F1-score, and ROC-AUC of 1.000, with its confusion matrix indicating no false positives or false negatives. Consequently, the conclusive findings designate the optimized XGBoost as the most effective model in this project.

Analysis and Discussion

Interpretation of Model Performance

The findings indicate that the three classifiers exhibited varying performance on the heart disease risk categorization test. Prior to optimization, XGBoost demonstrated the most robust baseline performance, succeeded by SVM and Decision Tree. This indicates that the clinical characteristics possessed valuable prediction patterns, although each program utilized those patterns distinctively. The Decision Tree yielded the least robust base result due to its propensity for inflexible feature splits, which may fail to encapsulate more intricate interactions among clinical factors. The SVM exhibited superior performance due to the scaled characteristics facilitating a more distinct decision boundary between low-risk and high-risk records. XGBoost excelled because to its sequential combination of many decision trees, effectively mitigating prior prediction errors. The foundational confusion matrices corroborated this conclusion, since XGBoost generated the least number of total mistakes prior to optimization.

Effect of Hyperparameter Tuning

The optimization of hyperparameters enhanced the performance of all three models, demonstrating that parameter selection significantly influenced outcomes. The modified Decision Tree diminished erroneous predictions and significantly outperformed its predecessor. The optimized SVM attained flawless recall, indicating that all genuine high-risk entries in the test set were accurately recognized. This is significant due to the substantial repercussions of false negatives in a health-risk classification endeavor. The optimized XGBoost attained peak performance, achieving 1.000 in accuracy, precision, recall, F1-score, and ROC-AUC. The confusion matrix indicated the absence of false positives and false negatives. This indicates that each test record was accurately classified inside the designated test set. This outcome should be regarded solely as test-set performance, not as evidence of flawless real-world clinical efficacy.

Comparison with Previous Studies

The findings correspond with previous research demonstrating that machine learning is superior in predicting cardiac disease. Mohan et al. (2019) utilized a hybrid Random Forest alongside a linear model; nevertheless, my experiment did not incorporate either methodology. This paper acknowledges the overall importance of machine learning, however it does not immediately correspond with the method in question. Ghosh et al. (2021) utilized integrated datasets, Relief and LASSO for feature selection, and ensemble methodologies. My experiment employed a singular prepared dataset and omitted feature selection, leading to a constrained methodology. Fitriyani et al. (2020) is more pertinent to my project as it utilized XGBoost; nevertheless, that study also incorporated DBSCAN and SMOTE-ENN, which are not present in my code. Li et al. (2020) support my use of SVM and Decision Tree; nevertheless, their study focused mostly on feature selection. The research validates my choice of Decision Tree, SVM, and XGBoost, while also suggesting that optimal results may depend on preprocessing, feature selection, class balancing, and validation design.

Limitations and Reliability

The primary constraint is that the optimized XGBoost model attained flawless performance on the test set. This is the most robust outcome, however it requires careful consideration. The dataset had 970 refined records, adequate for an academic research but constrained for clinical generalization. The dataset was sourced from Kaggle, and the original collector, country, collecting year, and clinical collection process were not explicitly specified. This constrains confidence in the dataset's representativeness. The `target_class` variable was derived from an existing decimal target score, perhaps simplifying the classification boundary for the models to learn. Moreover, no external hospital dataset was employed for validation purposes. Consequently, the ultimate model ought to be regarded as a data science classification model rather than a medical diagnostic system.

Answer to the Research Question

The research issue inquired if the classification efficacy of Decision Tree, SVM, and XGBoost differs when predicting low-risk and high-risk heart disease cohorts based on clinical features. The data indicate that performance fluctuated. XGBoost served as the most robust foundation model, while the optimized XGBoost emerged as the most formidable end model. The Support Vector Machine (SVM) exhibited commendable performance, particularly following optimization, whilst the Decision Tree offered a comprehensible baseline. The use of accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC rendered the evaluation more comprehensive than reliance on accuracy alone. This experiment demonstrates that machine learning can effectively classify the two risk groups in the supplied dataset; however, broader validation is necessary prior to clinical use, and expert medical evaluation is essential before implementation.

Conclusion

Project Summary

This project developed and evaluated machine learning models for risk of heart disease classification from clinical health features. I prepared the dataset by checking data quality, removing duplicate records, and transforming the original decimal target score to a binary `target_class` to distinguish between low-risk and high-risk groups. The final modelling dataset after pre-processing consisted of 970 records and 13 clinical input features.

Main Findings

3 supervised classification models were trained and tested . SVM, Decision Tree and XGBoost The accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC of each model were evaluated before and after hyperparameter tuning. The results indicated that the models were not performing equally. XGBoost reached the best base performance and tuned XGBoost reached the best final result, scoring 1.000 in all

evaluation metrics on the test dataset. Thus the research question was answered as the classification performance of Decision Tree, SVM and XGBoost was distinctly varied while predicting low risk and high risk heart disease groups.

Limitations and Future Work

The best result was given by the tuned XGBoost, but I have to be careful with the interpretation of this result. The dataset was downloaded from Kaggle and the original collection was not clearly mentioned and no external hospital dataset was used for validation. Therefore the final model should be viewed as a data science classification tool and not a medical diagnostic tool. In future work I plan to test the model on larger external datasets, apply feature selection methods and validate the performance using clinically verified data.

References

Fitriyani, N.L., Syafrudin, M., Alfian, G. and Rhee, J. (2020) 'HDPM: An Effective Heart Disease Prediction Model for a Clinical Decision Support System', *IEEE Access*, 8, pp. 133034–133050.

doi: 10.1109/ACCESS.2020.3010511. Available at:

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9144587&isnumber=8948470> (

Ghosh, P., Azam, S., Jonkman, M., Karim, A., Shamrat, F.M.J.M., Ignatious, E., Shultana, S., Beeravolu, A.R. and De Boer, F. (2021) 'Efficient Prediction of Cardiovascular Disease Using Machine Learning Algorithms With Relief and LASSO Feature Selection Techniques', *IEEE Access*, 9, pp. 19304–19326. doi: 10.1109/ACCESS.2021.3053759. Available at:

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9333574&isnumber=9312710>

Li, J.P., Haq, A.U., Din, S.U., Khan, J., Khan, A. and Saboor, A. (2020) 'Heart Disease Identification Method Using Machine Learning Classification in E-Healthcare', *IEEE Access*, 8, pp. 107562–107582. doi: 10.1109/ACCESS.2020.3001149. Available at:

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9112202&isnumber=8948470>

Mohan, S., Thirumalai, C. and Srivastava, G. (2019) 'Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques', *IEEE Access*, 7, pp. 81542–81554. doi: 10.1109/ACCESS.2019.2923707. Available at:

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8740989&isnumber=8600701>